

3/4 UNIT MATHEMATICS FORM VI

Time allowed: 2 hours (plus 5 minutes reading)

Exam date: 16th August, 1999

Instructions:

- All questions may be attempted.
- All questions are of equal value.
- Part marks are shown in boxes in the left margin.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection:

- Each question will be collected separately.
- Start each question in a new answer booklet.
- If you use a second booklet for a question, place it inside the first. Don't staple.
- Write your candidate number on each answer booklet.

QUESTION ONE (Start a new answer booklet)

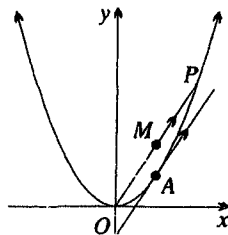
Marks

- 2** (a) Find and simplify the term in x^5 in the expansion of $(2 - x)^7$.
- 2** (b) Differentiate $e^{2x} \sin x$.
- 2** (c) Find the gradient of the tangent to $y = \sin^{-1} \frac{x}{2}$ at the point where $x = 1$.
- 2** (d) Solve $x^2 - x - 6 > 0$.
- 2** (e) Find, correct to the nearest minute, the acute angle between the lines $x - y + 3 = 0$ and $2x + y + 1 = 0$.
- 2** (f) Find:
- (i) $\int \frac{1 + e^x}{e^x} dx$,
- (ii) $\int \frac{e^x}{1 + e^x} dx$.

QUESTION TWO (Start a new answer booklet)

Marks

- 2** (a) Find the general solution of $\cos x = -\frac{1}{2}$.
- 2** (b) What are the coordinates of the focus of the parabola $(x + 3)^2 = 8(y - 1)$?
- 4** (c)



The point $P(2ap, ap^2)$ and the origin O lie on the parabola $x^2 = 4ay$. M is the mid-point of the chord OP .

- (i) Find the gradient of OP .
- (ii) Show that the tangent at a point $T(2at, at^2)$ on the parabola has gradient t .
- (iii) Hence find the point A on the parabola where the tangent is parallel with the chord OP , and show that A is equidistant from M and the x -axis.
- 4** (d) (i) Show $\alpha^3 + \beta^3 = (\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta)$.
- (ii) Given α and β are roots of the quadratic equation $x^2 + 3x - 2 = 0$, find the value of $\alpha^3 + \beta^3$ without finding the values of the roots.

QUESTION THREE (Start a new answer booklet)

Marks

- 4** (a) (i) Prove that $\sin^2 \theta = \frac{1}{2} - \frac{1}{2} \cos 2\theta$.
- (ii) Hence determine $\int_0^\pi \sin^2 \theta \, d\theta$.
- 4** (b) Use the substitution $u = 1 - x$ to help evaluate $\int_0^1 (1 + 3x)(1 - x)^7 \, dx$.
- 4** (c) (i) Write down a value of θ for which $\frac{1}{1 + \sin \theta}$ is undefined.
- (ii) Show that $\frac{1}{1 + \sin \theta} = \sec^2 \theta - \sec \theta \tan \theta$.
- (iii) Hence find $\int \frac{1}{1 + \sin \theta} \, d\theta$. [HINT: You may want to consult the list of standard integrals.]

QUESTION FOUR (Start a new answer booklet)

Marks

- 3** (a) (i) Use sigma notation to express $(1+x)^{2n}$ as a sum of powers of x .

(ii) Hence show that $\sum_{r=0}^{2n} {}^{2n}C_r \left(-\frac{1}{2}\right)^r = \left(\frac{1}{2}\right)^{2n}$.

(iii) Hence evaluate $\sum_{r=0}^{2n-1} {}^{2n}C_r \left(-\frac{1}{2}\right)^r$.

- 4** (b) (i) Expand $\left(x - \frac{1}{x}\right)^2$.

(ii) Show that $\left(x^2 + \frac{1}{x^2}\right)^{14} = \sum_{r=0}^{14} {}^{14}C_r x^{28-4r}$.

(iii) Hence show that the coefficient of x^6 in the expansion of $\left(x - \frac{1}{x}\right)^2 \left(x^2 + \frac{1}{x^2}\right)^{14}$ is equal to ${}^{15}C_6$.

- 5** (c) (i) An amount P is borrowed from a bank at an interest rate of R per month compounded monthly. At the end of each month, an instalment M is paid back to the bank. Let A_n be the amount owed at the end of the n^{th} month, after the instalment is paid. Show that:

$$A_n = P(1+R)^n - \frac{M((1+R)^n - 1)}{R}.$$

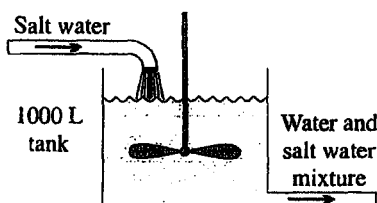
- (ii) A couple want to borrow \$20 000 from the bank, for a new car. After all charges are taken into account, the effective interest rate for the personal loan is 1.2% per month, compounded monthly, with the loan to be repaid over 5 years. The couple can only afford to make repayments of \$450 per month. Will the bank give them the loan? Justify your answer.

QUESTION SIX (Start a new answer booklet)

Marks

8

(a)



In the diagram above, a tank initially contains 1000 L of pure water. Salt water begins pouring into the tank from a pipe and a stirring blade ensures it is completely mixed with the pure water. A second pipe draws the water and salt water mixture off at the same rate, so that there is always a total of 1000 L in the tank.

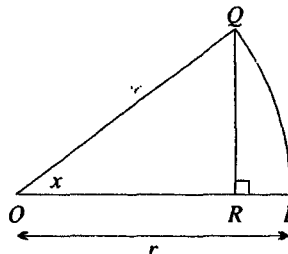
- (i) If the salt water entering the tank contains 2 grams of salt per litre and is flowing in at the constant rate of w L/min, how much salt is entering the tank per minute?
- (ii) If Q grams is the amount of salt in the tank at time t , how much salt is in 1 L at time t ?
- (iii) Hence write down the amount of salt leaving the tank per minute.
- (iv) Use the previous parts to show that $\frac{dQ}{dt} = -\frac{w}{1000}(Q - 2000)$.
- (v) Show that $Q = 2000 + Ae^{-\frac{wt}{1000}}$ is a solution of this differential equation.
- (vi) Determine the value of A .
- (vii) What happens to Q as $t \rightarrow \infty$?
- (viii) If there is 1 kg of salt in the tank after $5\frac{3}{4}$ hours, find w .

- 4 (b) A pupil investigated a differentiable function $f(x)$ and found the following information: $f(x)$ has its only zero at $x = -1$, $f(0) = 2$, $\lim_{x \rightarrow \infty} f(x) = 0$.
 - (i) Draw a graph of the possible shape of $f(x)$.
 - (ii) Use your graph to demonstrate that $f(x)$ must have an inflexion point to the right of $x = -1$.

QUESTION SEVEN (Start a new answer booklet)

Marks

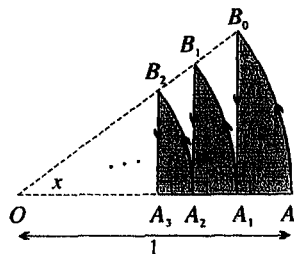
- 7** (a) (i) In the diagram on the right, PQ is the arc of a circle with radius r subtending an acute angle x at the centre O . R is the foot of the perpendicular from Q to the radius OP . Find lengths of the arc PQ and the interval QR in terms of x and r .



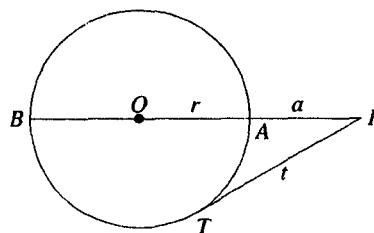
- (ii) An ant travels from A_0 to O along the saw-tooth path as shown in the diagram on the right. Show that the total distance y travelled by the ant is:

$$y = \frac{x + \sin x}{1 - \cos x}$$

- (iii) Given $0 < x \leq \frac{\pi}{2}$, use the derivative of y to find the value of x that gives the shortest such distance.

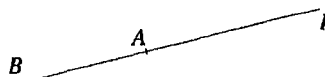


- 5** (b) (i) In the diagram, P is a point outside a circle with centre O and radius r . The secant PO cuts the circle at A and B respectively, and $PA = a$. PT is tangent to the circle at T and $PT = t$.



- (α) Give a reason why $t^2 = a(a + 2r)$.
 (β) Solve this equation for a and hence show the geometric mean of PA and PB is less than the arithmetic mean.
 NOTE: The geometric mean of a and b is $\sqrt{ab}$ and arithmetic mean is $\frac{a+b}{2}$.

- (ii) The diagram on the right shows the interval PAB . A circle is drawn to pass through A and B . A tangent is drawn from P to touch the circle at T . Find and describe the locus of T for all such circles and tangents.



DNW

$$1 \quad a) \quad \text{Term} = x^5 = {}^7C_5 \cdot 2^2 (-x)^5 \quad (1)$$

$$= -84x^5 \quad (1)$$

$$b) \quad y = e^{2x} \sin x$$

$$\frac{dy}{dx} = 2e^{2x} \sin x + e^{2x} \cos x \quad (1) + (1)$$

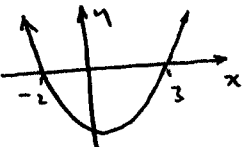
$$= e^{2x} (2\sin x + \cos x)$$

$$c) \quad \frac{dy}{dx} = \frac{1}{\sqrt{4-x^2}} \quad (1)$$

$$\text{so gradient at } x=1 \text{ is } \frac{1}{\sqrt{4-1}} = \frac{1}{\sqrt{3}} \quad (1)$$

$$d) \quad (x-3)(x+2) > 0$$

from the graph

$$x < -2 \text{ or } x > 3$$


(1)

(1)

$$e) \quad \text{Let } \phi \text{ be the angle}$$

$$\tan \phi = \left| \frac{1 - (-2)}{1 + 1 \times -2} \right| \quad (1)$$

$$= 3$$

$$\text{so } \phi = 71^\circ 34' \text{ (to nearest minute)} \quad (1)$$

$$f) \quad (i) \quad \int \frac{1+e^x}{e^x} dx = \int e^{-x} + 1 dx \quad (1)$$

$$= -e^{-x} + x + C$$

$$(ii) \quad \int \frac{e^x}{1+e^x} dx = \log(1+e^x) + C. \quad (1)$$

2, a) $\cos x = -\frac{1}{2}$

so x is in 2nd or 3rd quadrant

thus $x = \frac{2\pi}{3} + 2n\pi$ or $-\frac{2\pi}{3} + 2n\pi$

b) vertex is $(-3, 1)$, focal length $= 2$, axis vertical

so focus is $(-3, 3)$

c) (i) gradient $OP = \frac{ap^2}{2ap} = \frac{p}{2}$

(ii) $\frac{dy}{dx} = \frac{(dy/dt)}{(dx/dt)}$
 $= \frac{2at}{2a}$
 $= t$

(iii) thus at A parameter $t = \frac{p}{2}$

so $A = (ap, \frac{ap^2}{4})$

and $M = (ap, \frac{ap^2}{1})$

clearly y-coord of M is twice y-coord of A, as required

d) (i) Expand RHS or

$(\alpha + \beta)^3 = \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3$
 $= \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$

so $\alpha^3 + \beta^3 = (\alpha + \beta) [(\alpha + \beta)^2 - 3\alpha\beta]$

or ---

(ii) here $\alpha + \beta = -3$ and $\alpha\beta = -2$

so $\alpha^3 + \beta^3 = -3 [(-3)^2 - 3(-2)]$

$= -45$

$$\begin{aligned}
 3. \quad a) \quad (i) \quad RHS &= \frac{1}{2} (1 - \cos 2\theta) & (1) \\
 &= \frac{1}{2} (1 - \cos^2 \theta + \sin^2 \theta) & (1) \\
 &= \frac{1}{2} \cdot 2 \sin^2 \theta & (1) \\
 &= LHS \quad \#
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \int_0^\pi \sin^2 \theta \, d\theta &= \int_0^\pi \frac{1}{2} (1 - \cos 2\theta) \, d\theta & (1) \\
 &= \frac{1}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^\pi & (1) \\
 &= \frac{\pi}{2} & (1)
 \end{aligned}$$

$$\begin{aligned}
 b) \quad u &= 1-x & (1) \\
 \text{at } x=0 \quad u &= 1 \quad \text{and at } x=1 \quad u=0 & (1) \\
 x &= 1-u \\
 dx &= -du \\
 \text{so } \int_0^1 (1+3x)(1-x)^7 \, dx &= \int_1^0 (4-3u) u^7 \cdot (-du) & (1) \\
 &= \int_0^1 4u^7 - 3u^8 \, du & (1) \\
 &= \left[\frac{u^8}{8} - \frac{u^9}{9} \right]_0^1 & (1) \\
 &= \frac{1}{6} & (1)
 \end{aligned}$$

$$\begin{aligned}
 c) \quad (i) \quad \text{when } 1 + \sin \theta &= 0 & (1) \\
 \text{ie } \theta &= \frac{3\pi}{2} + 2n\pi.
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad LHS &= \frac{1}{1+\sin \theta} \cdot \frac{1-\sin \theta}{1-\sin \theta} & (1) \\
 &= \frac{1-\sin \theta}{\cos^2 \theta} & (1) \\
 &= \sec^2 \theta - \sec \theta \tan \theta \\
 &= RHS \quad \#
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad \int \frac{1}{1+\sin \theta} \, d\theta &= \int \sec^2 \theta - \sec \theta \tan \theta \, d\theta & (1) \\
 &= \tan \theta - \sec \theta + C
 \end{aligned}$$

$$4/ a) (i) (1+x)^{2n} = \sum_{r=0}^{2n} {}^{2n}C_r x^r \quad (1)$$

$$(ii) \text{ at } x = -\frac{1}{2} \quad (1)$$

$$\left(\frac{1}{2}\right)^{2n} = \sum_{r=0}^{2n} {}^{2n}C_r \left(-\frac{1}{2}\right)^r$$

$$(iii) \left(\frac{1}{2}\right)^{2n} = \sum_{r=0}^{2n-1} {}^{2n}C_r \left(-\frac{1}{2}\right)^r + \left(\frac{1}{2}\right)^{2n} \text{ from part (ii)}$$

$$\text{Thus } \sum_{r=0}^{2n-1} {}^{2n}C_r \left(-\frac{1}{2}\right)^r = 0 \quad (1)$$

$$b) (i) x^2 - 2 + \frac{1}{x^2} \quad (1)$$

$$(ii) \left(x^2 + \frac{1}{x^2}\right)^{14} = \sum_{r=0}^{14} {}^{14}C_r (x^2)^{14-r} (x^{-2})^r \quad (1)$$

$$= \sum_{r=0}^{14} {}^{14}C_r x^{28-4r}$$

$$(iii) \left(x - \frac{1}{x}\right)^4 \left(x^2 + \frac{1}{x^2}\right)^{14} = \left(x^2 - 2 + \frac{1}{x^2}\right) \sum_{r=0}^{14} {}^{14}C_r x^{28-4r}$$

$$= \sum_{r=0}^{14} {}^{14}C_r x^{30-4r} - 2 \sum_{r=0}^{14} {}^{14}C_r x^{28-4r} + \sum_{r=0}^{14} {}^{14}C_r x^{26-4r}$$

$$x^6 \text{ term comes from } r=6 \text{ in 1st sum} \quad (1)$$

$$\text{and } r=5 \text{ in last sum}$$

$$\text{so coeff of } x^6 = {}^{14}C_6 + {}^{14}C_5 \quad (1)$$

$$= {}^{15}C_6 \quad \text{by the recurrence relation (Pascal's } \Delta)$$

$$c) (i) A_0 = P \quad (1)$$

$$A_1 = P(1+R) - M$$

$$A_2 = P(1+R)^2 - M(1+R) - M$$

$\vdots$

$$A_n = P(1+R)^n - M[(1+R)^{n-1} + \dots + (1+R) + 1] \quad (1)$$

$$= P(1+R)^n - \frac{M[(1+R)^n - 1]}{R}$$

$$(ii) \text{ Here } A_n = 0 \text{ so } P = \frac{M[(1+R)^n - 1]}{R(1+R)} \quad (1)$$

$$\text{and } M = 450, R = 0.012, n = 60 \quad (1)$$

$$\text{for which } P \approx 19168 < 20000 \quad (1)$$

The bank will not give them the loan

5/ a) (i) $\ddot{x} = -3 \sin 3t + 6 \cos 3t$
 $\ddot{x} = -9 \cos 3t - 18 \sin 3t$
 $= -3^2 (\cos 3t + 2 \sin 3t)$

①

and $\lambda = 3$

(ii) $r^2 = 1^2 + 2^2 = 5$

so $\sin \alpha = \frac{1}{\sqrt{5}}$ and $\cos \alpha = \frac{2}{\sqrt{5}}$

①

hence $r = \sqrt{5}$, $\alpha = \sin^{-1}(\frac{1}{\sqrt{5}})$

$(\approx 0.46 \text{ rads})$

①

(iii) $r \sin(3t + \alpha) = 2$

so $3t = \sin^{-1}(\frac{2}{\sqrt{5}}) - \sin^{-1}(\frac{1}{\sqrt{5}})$

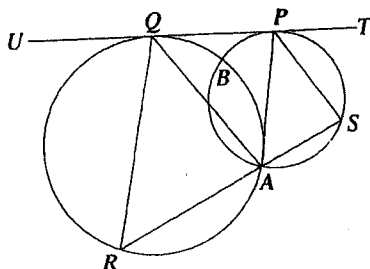
①

$t = \frac{1}{3} [\sin^{-1}(\frac{2}{\sqrt{5}}) - \sin^{-1}(\frac{1}{\sqrt{5}})]$

≈ 0.2 to 1 dec. pl.

①

b)



(i) exterior angle of cyclic quadrilateral.

①

(ii) $\angle QAR = \angle UQR$ (angle in alternate segment)
 $= \angle PSA$

①

hence $QA \parallel PS$ (corresponding angles equal)

①

(iii) $\angle PAS = \angle TPS$ (angle in alt segment)

$= \angle QRA$ (exterior angle of cyclic quadrilateral)

①

hence in $\triangle QRA$ and $\triangle PAS$

$\angle QAR = \angle PSA$ proven

$\angle PAS = \angle QRA$ proven

①

hence $\triangle QRA \parallel \triangle PAS$ (AA)

①

c) (i) 4a

(ii) in the right hand graph, the focal length is longer but the latus rectum is shorter.

①

6/ a (i) 2ω g/min

①

(ii) $\frac{Q}{1000}$ g/L

①

(iii) $\frac{Q\omega}{1000}$ g/min

①

(iv) $\frac{dQ}{dt} = \text{inflow} - \text{outflow}$
 $= 2\omega - \frac{Q\omega}{1000}$
 $= -\frac{\omega}{1000} (Q - 2000)$

①

(v) LHS = $-\frac{\omega}{1000} \cdot A e^{-\omega t/1000}$

RHS = $-\frac{\omega}{1000} (2000 + A e^{-\omega t/1000} - 2000)$

①

$= -\frac{\omega}{1000} A e^{-\omega t/1000}$

$= \text{LHS} \quad \#$

(vi) at $t=0$ $Q=0$ so $A = -2000$

①

and $Q = 2000(1 - e^{-\omega t/1000})$

(vii) as $t \rightarrow \infty$, $e^{-\omega t/1000} \rightarrow 0$ hence $Q \rightarrow 2000$

①

(viii) $1000 = 2000(1 - e^{-\omega 345/1000})$

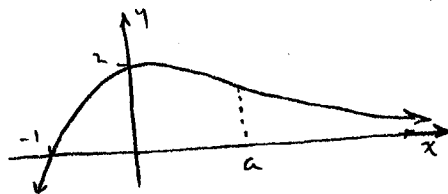
$e^{\omega 345/1000} = 2$

$\omega = \frac{1000}{345} \log 2$

①

$(\div 2 \text{ L/min.})$

b) (i)



(other graphs are possible)

①

①

(ii) for $x < a$ $f(x)$ is concave down

①

for $x > a$ $f(x)$ is concave up

hence $f(x)$ changes concavity and there is an inflection point.

①

More precisely, for the curve to rise from x and return to the x -axis it must be concave down in some domain. Also $f(x)$ must

7 a) (i) $PQ = rx$

$QR = r \sin x$

(1)

(ii) along each tooth of radius r the ant travels $r(x + \sin x)$
each successive tooth has radius $\cos x$ times the previous

(1)

so $y = (x + \sin x) + \cos x (x + \sin x) + \cos^2 x (x + \sin x) + \dots$

(1)

$$= \frac{x + \sin x}{1 - \cos x}$$

(1)

(iii) $y' = \frac{(1 - \cos x)(1 + \cos x) - (x + \sin x)(\sin x)}{(1 - \cos x)^2}$

$$= \frac{\sin^2 x - x \sin x - \sin^2 x}{(1 - \cos x)^2}$$

$$= \frac{-x \sin x}{(1 - \cos x)^2} < 0 \text{ for } 0 < x \leq \frac{\pi}{2}$$

(1)

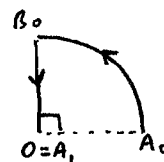
ie y is decreasing so min is at right

(1)

hand end pt

$$y\left(\frac{\pi}{2}\right) = \frac{\frac{\pi}{2} + 1}{1 - 0} = \frac{\pi}{2} + 1$$

(1)



b) (i) (α) the square of the tangent is equal to
the product of the intercepts of the secant

(1)

(β) $a^2 + 2ar + r^2 = 0$

$$a^2 + 2ar + r^2 = t^2 + r^2$$

$$(a+r)^2 = t^2 + r^2$$

$$a+r = \sqrt{t^2 + r^2}$$

$$\geq t$$

(1)

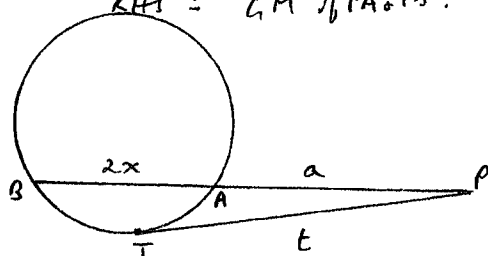
with equality then $r=0$.

LHS is AM of PA_0, PB

RHS is GM of PA_0, PB .

(1)

(ii)



$$t^2 = a(a+2x) \text{ so } t \text{ is constant}$$

so locus is the circle centre P radius t

(1)